Answers and talk notes

Math is Explaining and Sharing

What your student is learning

Tonight we build a math argument. We use volume — the space inside a box — to defend a recommendation with numbers instead of opinions.

Answers

Ask your student to explain before you look.

1. **B.** Cubic units. Cubic units. Volume counts the unit cubes that fill the space inside a box.
2. **A.** 24 cubic inches. 2 × 3 × 4 = 24 cubic inches.
3. **C.** 5 × 2 × 6. Volume is length × width × height, so 5 × 2 × 6 = 60 cubic inches.
4. **D.** 60 cubic inches. 6 × 5 × 2 = 60 cubic inches.
5. **B.** Box B holds more, because 120 is greater than 90. Box B holds more, because 120 is greater than 90.

Talk about it

Which box holds more, and what numbers prove it?

Home mission

Find a cereal box and a snack box. Measure each side in inches with a ruler. Multiply to find each volume. Can your student say which box holds more and finish the sentence with "because"?

When they are stuck, ask

- What are the three measurements of this box?
- Which total is bigger — and how do you know?
- Can you say your answer starting with "because"?

Try not to say

- Just look at it — the bigger one is obvious.
- You don't need to explain your answer.
- Add the three measurements together.